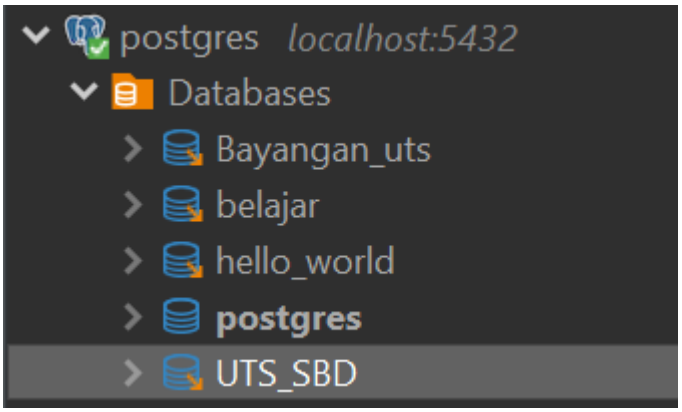
UTS SBD

Nama : Haesam Abdullah
NIM : 1251420101

DOKUMENTASI HASIL PENGERJAAN

1. Buatkan Data Base Terlebih dahulu



2. Lalu kita buatkan masing masing tabel kolom yang di perlukan dalam pengerjaan soal berikut;
   - Buat kolom tabel hotel

```
CREATE TABLE Hotel (
    hotelNo VARCHAR(10) NOT NULL,
    hotelName VARCHAR(50) NOT NULL,
    city VARCHAR(50) NOT NULL,
    PRIMARY KEY (hotelNo)
);
```

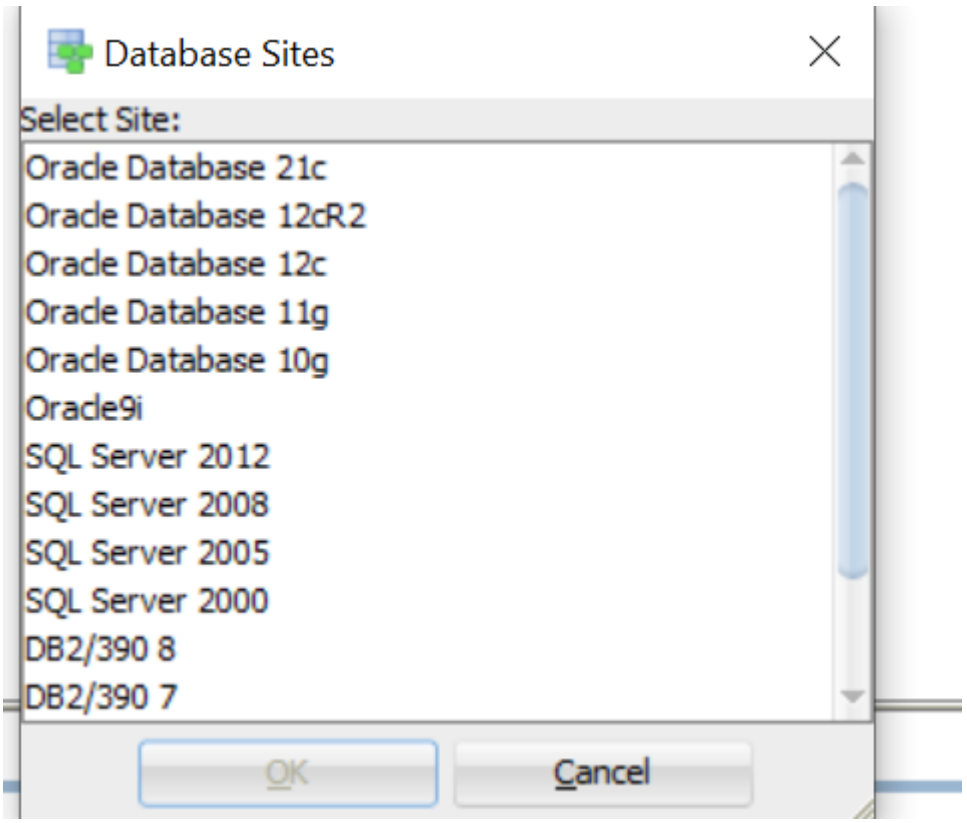
   - Buat kolom tabel guest, booking , room

```sql
CREATE TABLE Room (
    roomNo INT NOT NULL,
    hotelNo VARCHAR(10) NOT NULL,
    type VARCHAR(10) DEFAULT 'Single' NOT NULL,
    price DECIMAL(5, 2) NOT NULL,
    PRIMARY KEY (roomNo, hotelNo),
    FOREIGN KEY (hotelNo) REFERENCES Hotel(hotelNo),
    CONSTRAINT chk_room_type CHECK (type IN ('Single', 'Double', 'Family')),
    CONSTRAINT chk_room_price CHECK (price BETWEEN 10 AND 100),
    CONSTRAINT chk_room_no CHECK (roomNo BETWEEN 1 AND 100)
);

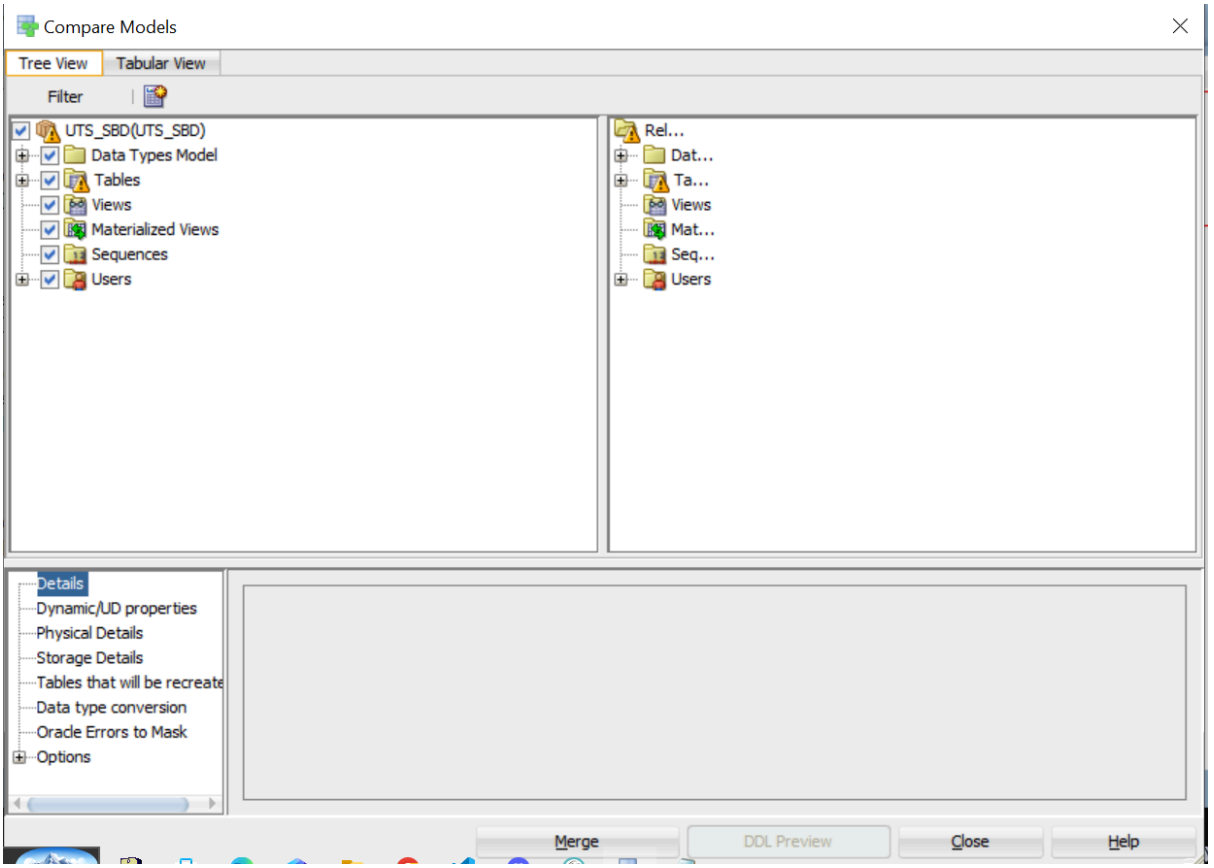
CREATE TABLE Guest (
    guestNo VARCHAR(10) NOT NULL,
    guestName VARCHAR(50) NOT NULL,
    guestAddress VARCHAR(100) NOT NULL,
    PRIMARY KEY (guestNo)
);

CREATE TABLE Booking (
    hotelNo VARCHAR(10) NOT NULL,
    guestNo VARCHAR(10) NOT NULL,
    dateFrom DATE NOT NULL,
    dateTo DATE NOT NULL,
    roomNo INT NOT NULL,
    PRIMARY KEY (hotelNo, guestNo, dateFrom),
    FOREIGN KEY (hotelNo) REFERENCES Hotel(hotelNo),
    FOREIGN KEY (guestNo) REFERENCES Guest(guestNo),
    FOREIGN KEY (roomNo, hotelNo) REFERENCES Room(roomNo, hotelNo)
);
```

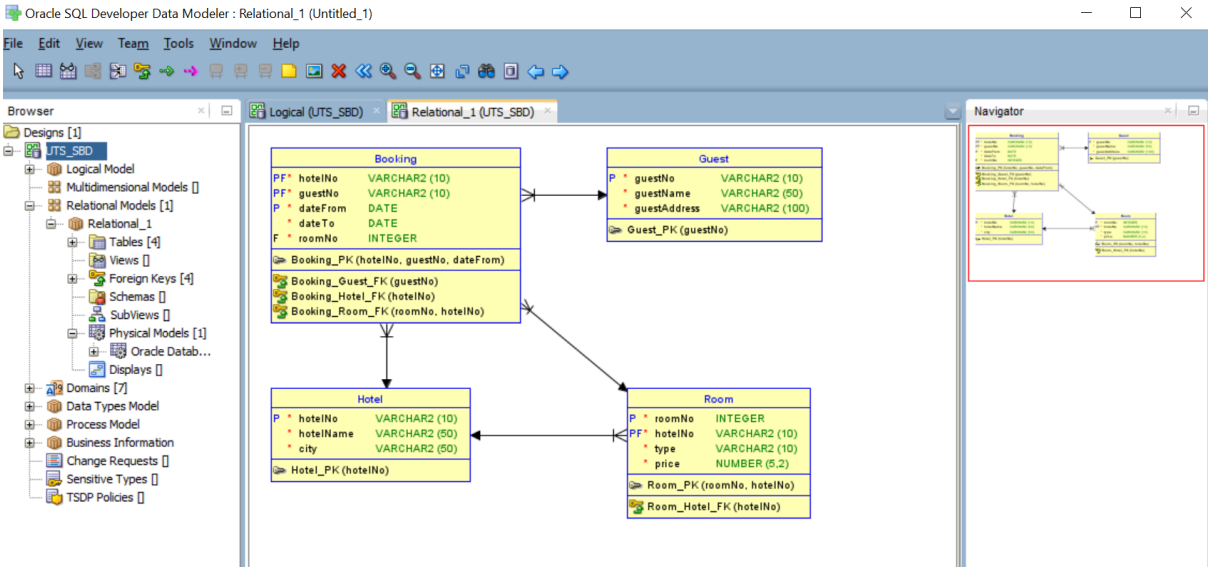
3. Setelah di buatkan file database hotel tersebut maka kita eksport ke dalam file .txt atau .sql untuk kita lakukan import ke Oracle SQL developer data model
4. Generate DDL PostgreSQL
    - Pada Physical Model, pilih File → Export → DDL File→ Generate DDL.
5. Kita masuk ke bagian Oracle SQL developer data model
   Setelah import file .sql tdi kita pilih ini menggunakan oracle databse 12c



6. Lalu kita Merge pada compire model



7. Maka akan menghasilkan tabel relasiny atau ERD



MENGGUNAKAN TOOLS ORACLE

1. Buka notepad untuk membuat catatan sql
2. Siapkan masing masing tabel yg harus kita buatkan
    - Berikut isi query yg perlu ada

```sql
CREATE TABLE Branch (
    branchNo VARCHAR(10) NOT NULL,
    mgrStaffNo VARCHAR(10),
    street VARCHAR(50),
    city VARCHAR(30),
    postcode VARCHAR(15),
    CONSTRAINT Branch_PK PRIMARY KEY (branchNo)
);

-- 2. Tabel Staff
CREATE TABLE Staff (
    staffNo VARCHAR(10) NOT NULL,
    supervisorStaffNo VARCHAR(10),
    branchNo VARCHAR(10) NOT NULL,
    fName VARCHAR(30),
    lName VARCHAR(30),
    position VARCHAR(20),
    sex CHAR(1),
    DOB DATE,
    salary NUMERIC(10, 2),
    CONSTRAINT Staff_PK PRIMARY KEY (staffNo),
    CONSTRAINT Staff_Supervisor_FK FOREIGN KEY (supervisorStaffNo) REFERENCES Staff (staffNo),
    CONSTRAINT Staff_Branch_FK FOREIGN KEY (branchNo) REFERENCES Branch (branchNo)
);

-- Menambahkan Foreign Key Manages dari Branch ke Staff
ALTER TABLE Branch
    ADD CONSTRAINT Branch_MgrStaff_FK FOREIGN KEY (mgrStaffNo) REFERENCES Staff (staffNo);

-- 3. Tabel Manager
CREATE TABLE Manager (
    staffNo VARCHAR(10) NOT NULL,
    mgrStartDate DATE,
    bonus NUMERIC(10, 2),
    CONSTRAINT Manager_PK PRIMARY KEY (staffNo),
    CONSTRAINT Manager_Staff_FK FOREIGN KEY (staffNo) REFERENCES Staff (staffNo)
);

-- 4. Tabel Telephone
CREATE TABLE Telephone (
    telNo VARCHAR(20) NOT NULL,
    branchNo VARCHAR(10) NOT NULL,
    CONSTRAINT Telephone_PK PRIMARY KEY (telNo),
    CONSTRAINT Telephone_Branch_FK FOREIGN KEY (branchNo) REFERENCES Branch (branchNo)
);

-- 5. Tabel Client
CREATE TABLE Client (
    clientNo VARCHAR(10) NOT NULL,
    fName VARCHAR(30),
    lName VARCHAR(30),
    telNo VARCHAR(20),
    prefType VARCHAR(20),
    maxRent NUMERIC(10, 2),
    eMail VARCHAR(50),
    CONSTRAINT Client_PK PRIMARY KEY (clientNo)
);
```

```sql
-- 6. Tabel Registration
CREATE TABLE Registration (
    clientNo VARCHAR(10) NOT NULL,
    branchNo VARCHAR(10) NOT NULL,
    staffNo VARCHAR(10) NOT NULL,
    dateJoined DATE,
    CONSTRAINT Registration_PK PRIMARY KEY (clientNo),
    CONSTRAINT Registration_Client_FK FOREIGN KEY (clientNo) REFERENCES Client (clientNo),
    CONSTRAINT Registration_Branch_FK FOREIGN KEY (branchNo) REFERENCES Branch (branchNo),
    CONSTRAINT Registration_Staff_FK FOREIGN KEY (staffNo) REFERENCES Staff (staffNo)
);

-- 7. Tabel PrivateOwner (POwns PropertyForRent)
CREATE TABLE PrivateOwner (
    ownerNo VARCHAR(10) NOT NULL,
    fName VARCHAR(30),
    lName VARCHAR(30),
    address VARCHAR(100),
    telNo VARCHAR(20),
    eMail VARCHAR(50),
    CONSTRAINT PrivateOwner_PK PRIMARY KEY (ownerNo)
);

-- 8. Tabel BusinessOwner (BOwns PropertyForRent)
CREATE TABLE BusinessOwner (
    ownerNo VARCHAR(10) NOT NULL,
    bName VARCHAR(50),
    bType VARCHAR(30),
    address VARCHAR(100),
    telNo VARCHAR(20),
    eMail VARCHAR(50),
    contactName VARCHAR(50),
    CONSTRAINT BusinessOwner_PK PRIMARY KEY (ownerNo)
);

-- 9. Tabel PropertyForRent (Oversees Staff, Offers Branch, POwns/BOwns Owner)
CREATE TABLE PropertyForRent (
    propertyNo VARCHAR(10) NOT NULL,
    ownerNo VARCHAR(10) NOT NULL,
    staffNo VARCHAR(10),
    branchNo VARCHAR(10) NOT NULL,
    street VARCHAR(50),
    city VARCHAR(30),
    postcode VARCHAR(15),
    type VARCHAR(20),
    rooms INTEGER,
    rent NUMERIC(10, 2),
    CONSTRAINT PropertyForRent_PK PRIMARY KEY (propertyNo),
    CONSTRAINT Property_Staff_FK FOREIGN KEY (staffNo) REFERENCES Staff (staffNo),
    CONSTRAINT Property_Branch_FK FOREIGN KEY (branchNo) REFERENCES Branch (branchNo),
    CONSTRAINT Property_PrivateOwner_FK FOREIGN KEY (ownerNo) REFERENCES PrivateOwner (ownerNo),
    CONSTRAINT Property_BusinessOwner_FK FOREIGN KEY (ownerNo) REFERENCES BusinessOwner (ownerNo)
);
```

-

```sql
-- 10. Tabel Lease
CREATE TABLE Lease (
    leaseNo VARCHAR(10) NOT NULL,
    clientNo VARCHAR(10) NOT NULL,
    propertyNo VARCHAR(10) NOT NULL,
    rent NUMERIC(10, 2),
    paymentMethod VARCHAR(20),
    deposit NUMERIC(10, 2),
    paid CHAR(1),
    rentStart DATE,
    rentFinish DATE,
    CONSTRAINT Lease_PK PRIMARY KEY (leaseNo),
    CONSTRAINT Lease_Client_FK FOREIGN KEY (clientNo) REFERENCES Client (clientNo),
    CONSTRAINT Lease_Property_FK FOREIGN KEY (propertyNo) REFERENCES PropertyForRent (propertyNo)
);

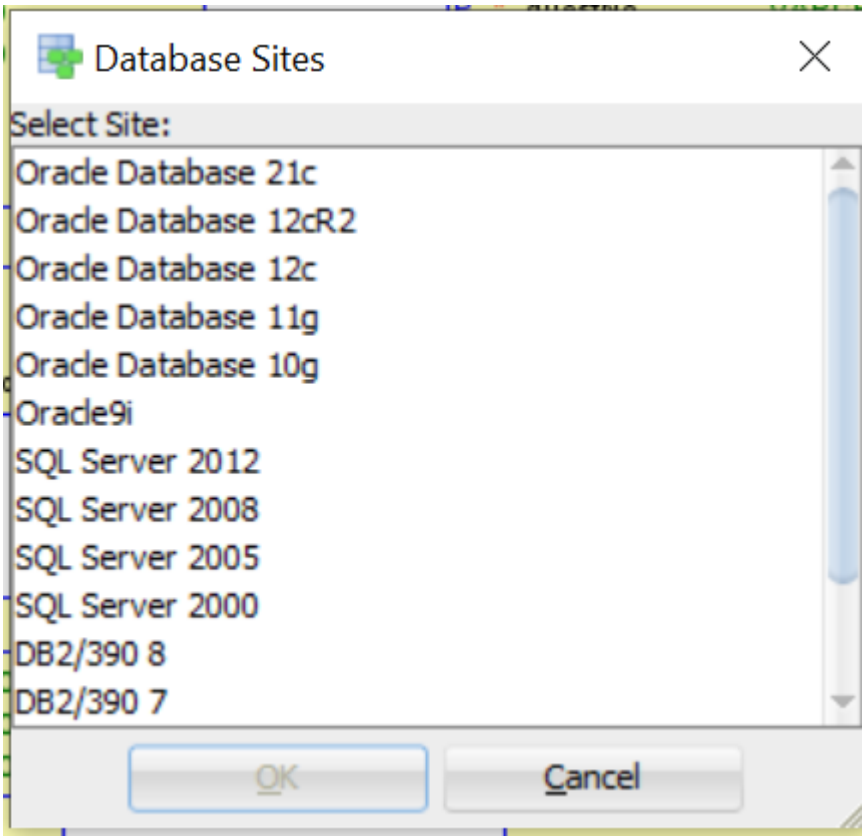
-- 11. Tabel Viewing
CREATE TABLE Viewing (
    clientNo VARCHAR(10) NOT NULL,
    propertyNo VARCHAR(10) NOT NULL,
    viewDate DATE NOT NULL,
    comment VARCHAR(200),
    CONSTRAINT Viewing_PK PRIMARY KEY (clientNo, propertyNo),
    CONSTRAINT Viewing_Client_FK FOREIGN KEY (clientNo) REFERENCES Client (clientNo),
    CONSTRAINT Viewing_Property_FK FOREIGN KEY (propertyNo) REFERENCES PropertyForRent (propertyNo)
);

-- 12. Tabel Newspaper
CREATE TABLE Newspaper (
    newspaperName VARCHAR(50) NOT NULL,
    address VARCHAR(100),
    telNo VARCHAR(20),
    contactName VARCHAR(50),
    CONSTRAINT Newspaper_PK PRIMARY KEY (newspaperName)
);

-- 13. Tabel Advert
CREATE TABLE Advert (
    propertyNo VARCHAR(10) NOT NULL,
    newspaperName VARCHAR(50) NOT NULL,
    dateAdvert DATE NOT NULL,
    cost NUMERIC(10, 2),
    CONSTRAINT Advert_PK PRIMARY KEY (propertyNo, newspaperName, dateAdvert),
    CONSTRAINT Advert_Property_FK FOREIGN KEY (propertyNo) REFERENCES PropertyForRent (propertyNo),
    CONSTRAINT Advert_Newspaper_FK FOREIGN KEY (newspaperName) REFERENCES Newspaper (newspaperName)
);
```

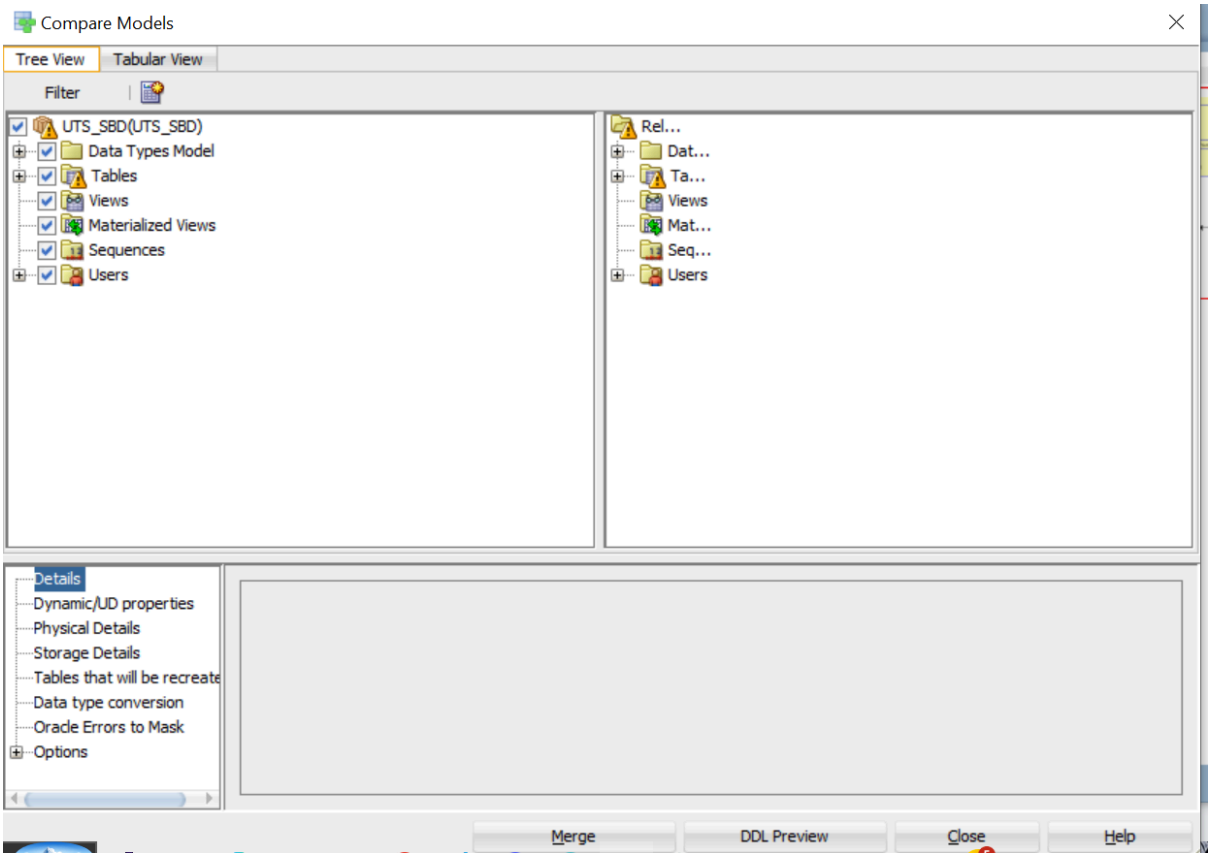
-

3. Lalu import file tersebut ke Oracle

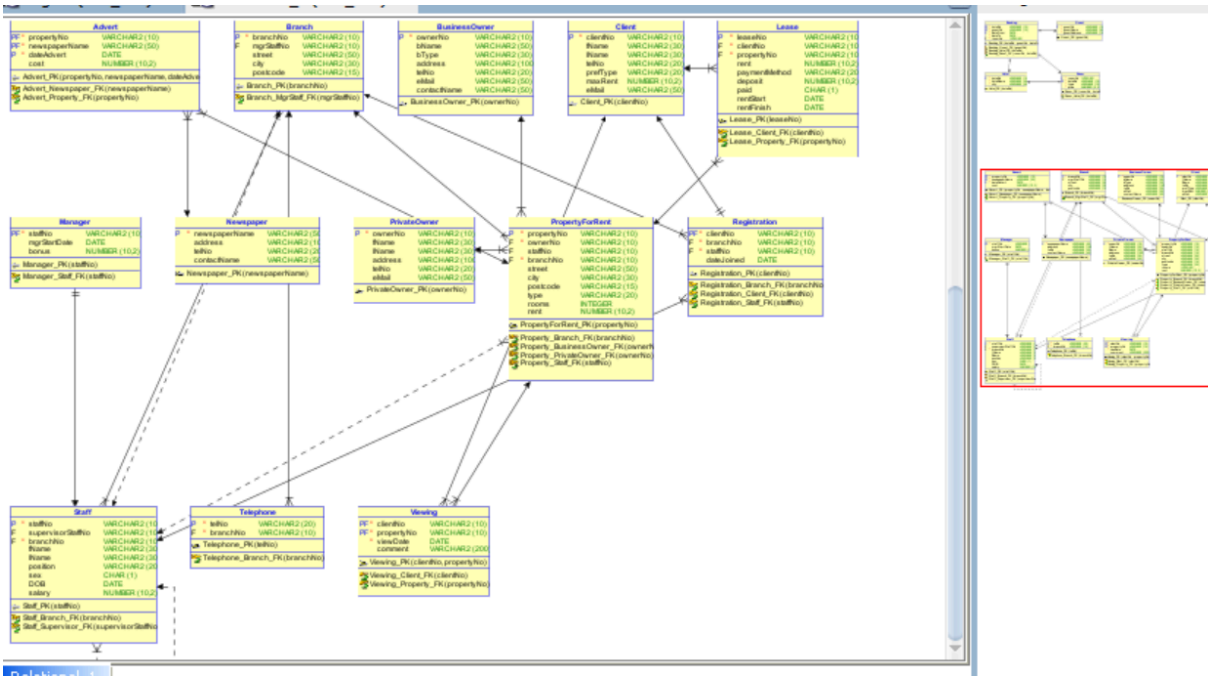


   - Pada bagian ini pilih pada oracle 12c

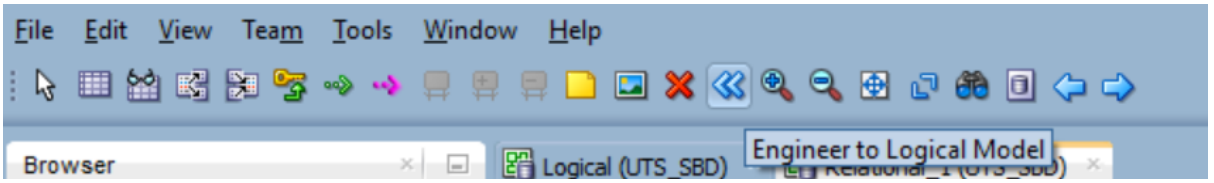
4. Generate DDL PostgreSQL
   - Pada Physical Model, pilih File → Export → DDL File→ Generate DDL.
5. Lalu Kita merge



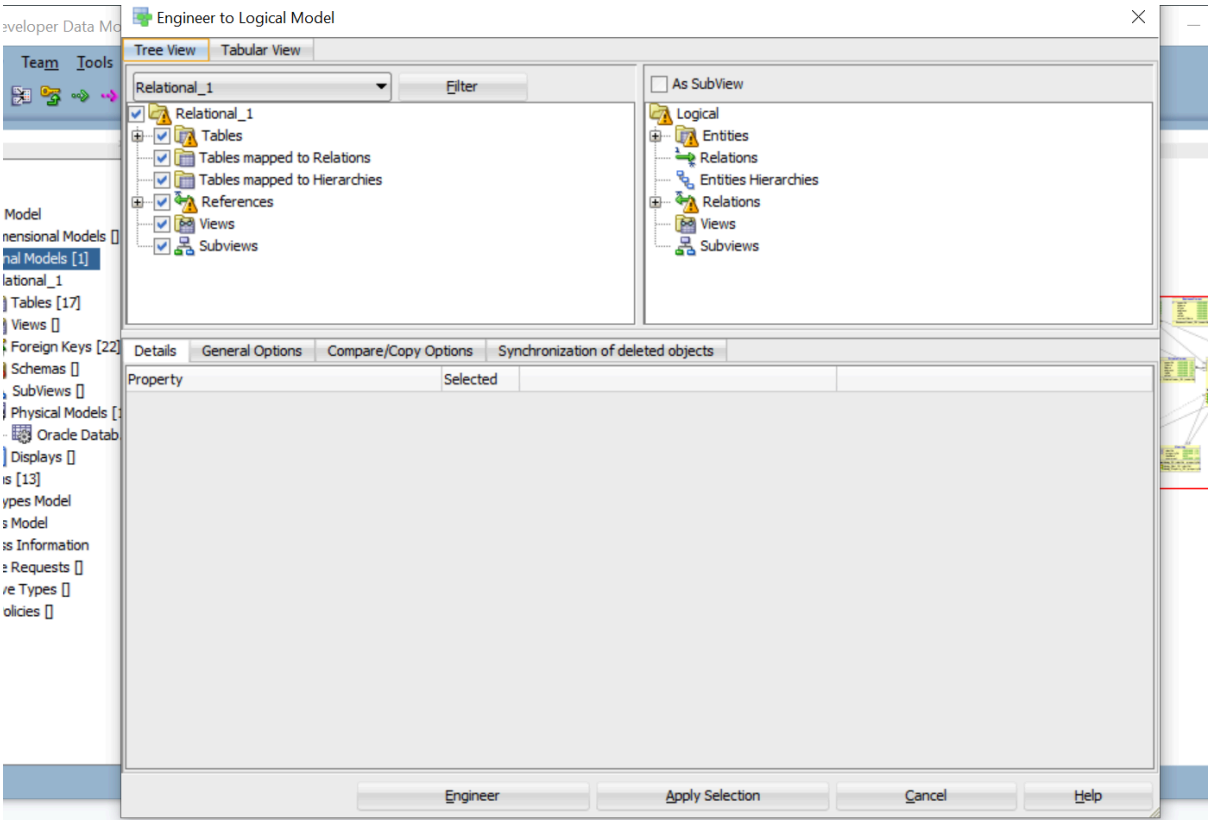
6. Maka akan menghasilkan relasi tersebut



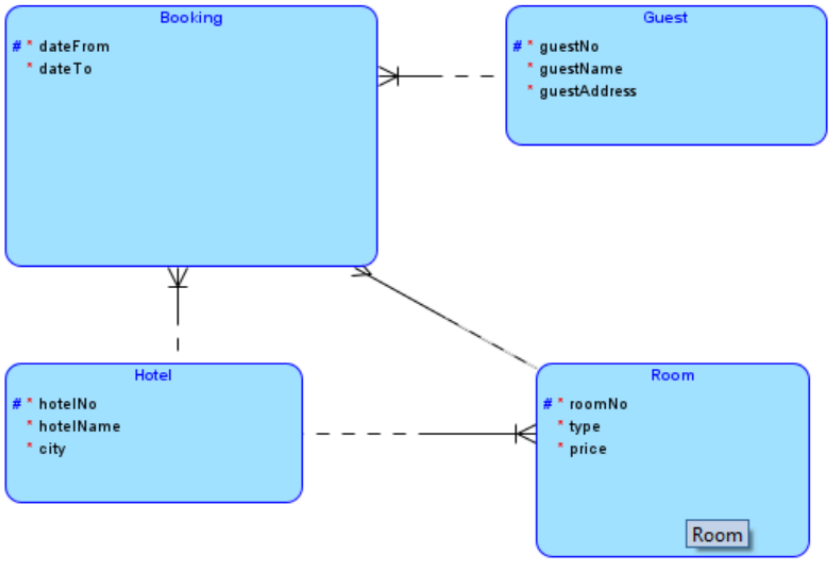
7. Langkah selanjutnya membuat phsyical model pada oracle
   Ikuti arahann tersebut pada gambar ikon ikon panah



8. Setelah itu klik pada bagian Enggineer model
    - Klik Enggineer



9. Maka masing masing akan menghasilkan tabel relasi yg berwarna biru seperti ini mulai dari sql hotel tadi, dan dreamrooms



10. Setelah jadi logic phsicalnya kita save aja ke dalam image

